

Predicting Hit Songs Using Repeated Chorus

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Problem statement and scope

Can 15-second chorus audio features distinguish songs with greater chart success? This project implements the chorus-based classification idea in Eric Liu's 2021 CS229 report [1]. It compares six model families and tests whether learned patterns transfer to unfamiliar artists.

The available public dataset uses a **year-end-hit proxy**: label 1 contains source year-end Hot 100 selections, while label 0 contains other songs sampled from weekly Hot 100 charts. Both classes can have charted. This differs from the reference paper's charted versus uncharted definition and is not an exact replication.

Dataset and audio features

We reuse Antonios Malak's Apache-2.0 feature dataset [2], pinned to commit 838e76f. It contains **751 songs, 71 artist names, and 518 numerical features**: 366 label-1 and 385 label-0 songs. The upstream collection spans 2006-2021. The local audit found no missing feature values, constant columns, or exact duplicate feature vectors. We remove audio paths and exclude all metadata from model inputs.

Feature family	Channels	Summary values
Chroma STFT, CQT, CENS	12 each	252 total
MFCC (timbre)	20	140
Spectral contrast; tonnetz	7; 6	49; 42
RMS, centroid, bandwidth, rolloff, zero-crossing rate	1 each	35 total
Total	74	518

Each channel contributes skewness, minimum, maximum, standard deviation, mean, median, and kurtosis. The source states that pychorus selected 15-second excerpts. Original recordings are not bundled, so the extraction and label assignments cannot be fully re-audited here.

Approach and implementation

A fixed artist-grouped split allocates **597 songs from 52 artists** to training and **154 songs from 19 different artists** to testing. Five artist-disjoint folds inside training select hyperparameters and the model by mean balanced accuracy. The seed is 42. No test scores guide selection.

A scikit-learn Pipeline fits median imputation, variance filtering, and standardization within each training fold. PCA retains 95% of training variance for logistic regression, LDA, three SVM kernels, and a neural network. Random forest and gradient boosting retain the original features. A majority-class classifier provides a baseline. The saved model remains fitted on training data only.

Python modules cover data checks, training, evaluation, inference, and audio processing. A Streamlit interface supports held-out song prediction, results inspection, and experimental uploaded-audio inference. A notebook and README explain the workflow.

Measured results and interpretation

Balanced accuracy averages recall across both labels. F1 refers to the year-end-hit class. Every test score below uses the same unseen-artist holdout.

Model	CV BA	Test BA	Accuracy	F1	AUC
Majority baseline	0.500	0.500	0.506	0.000	0.500
Logistic regression	0.523	0.525	0.526	0.503	0.546
LDA	0.528	0.526	0.526	0.523	0.525
Linear SVM	0.496	0.552	0.552	0.537	0.559
RBF SVM	0.528	0.505	0.506	0.449	0.533
Polynomial SVM *	0.536	0.466	0.468	0.406	0.456
Random forest	0.533	0.524	0.526	0.459	0.547
Gradient boosting	0.518	0.530	0.532	0.419	0.532
Neural network	0.499	0.527	0.526	0.552	0.514

* Selected using training cross-validation. Other test results are descriptive comparisons, not a reason to switch models after viewing the holdout.

Conclusion

The selected Polynomial SVM achieved **46.6% test balanced accuracy** and 0.406 F1, compared with 50.0% balanced accuracy for the baseline. The approximate 95% interval is 38.7%-55.8%, obtained by resampling whole test artists 2,000 times. It includes 50%. **This experiment provides no reliable evidence that these chorus features predict year-end-hit status for unfamiliar artists.** The result does not establish that chorus quality causes popularity.

Limitations and improvements

The dataset is small and historically sampled, and its labels are a proxy inherited from upstream collection code. Artist grouping uses provided names and cannot fully resolve collaborations. Original audio is unavailable, so feature-to-recording correctness remains unverified. The upload demo uses a new repetition heuristic or manual selection with current librosa, rather than exactly reproducing the old extraction environment.

A stronger follow-up would independently verify chart labels, collect licensed recordings for charted and uncharted tracks under a fixed time window, re-extract every clip consistently, and evaluate on later release dates. Further model tuning should use a new evaluation split or nested cross-validation.

Demonstration and reproducibility

Run the local app, choose a held-out song, compare prediction with its dataset label, then inspect the results table and error analysis. Uploaded audio can demonstrate segment selection and feature extraction, but its predictions are exploratory. The repository includes a pinned data source, dependency versions, split manifest, all CV trials, test predictions, tests, and the training-only model.

References

- [1] Eric Liu. *Predicting Hit Songs Using Repeated Chorus*, CS229, 2021. [Stanford report](#).
- [2] Antonios Malak. *Predicting-Hit-Songs-Using-Repeated-Chorus*, commit 838e76f. [Dataset and collection notebooks](#). Apache-2.0.
- [3] Scikit-learn documentation, *Common pitfalls and recommended practices*. [Data leakage and pipelines](#).